EVERY ABSOLUTELY SIMPLE LIE ALGEBRA OF TYPE E8, F4 OR G2 OVER Q IS R-UNIVERSAL

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned publisher source sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The classification of absolutely simple Lie algebras over Q by type, and the fact that types E8, F4 and G2 have no outer automorphisms and trivial centre. Used inside the counted proof at source lines 507-509; the classification frame is set up in the section the counted statement belongs to (source lines 378-400). Makes the star-action trivial on these types, which is the hypothesis of the closing step.
- The criterion for R-universality through the Tits algebras being split. The author's remark labelled OnlyZStar at source lines 461-468 and the author's equivalence labelled RunivIff at source lines 415-436, both invoked inside the counted proof. Turns the triviality of the star-action into the conclusion that every Tits algebra is Q itself, hence the algebra is R-universal.

Source reference. Original remark 2.4: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 462–462. Navigation pointer only: the counted proof invokes this remark in its second sentence; its content is declared as a named standard prerequisite and is not reproduced.

Source reference. Original lemma 2.1: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 416–416. Navigation pointer

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only: the counted proof invokes the author's equivalence in its last sentence; the equivalence has its own card in this lane.

Source reference. Original definition 3: sources/algebra-fl02-a-cohomological-proof-that-real-re/pinned-original-source.tex, source lines 428–428. Navigation pointer only: it is the third labelled item of that equivalence, cited by the counted proof once.

Lemma 0.1. *Every absolutely simple Lie algebra of type E_8, F_4, or G_2 over $\mathbb{Q}$ is $\mathbb{R}$-universal.*

Proof. These types have no outer automorphisms, so the $*$-action must be trivial. Furthermore, simple groups of these types have trivial center. Therefore, it is immediate from Remark 2.4 that $D_{\mathfrak{g},\mathbb{Q}}(\lambda) = \mathbb{Q}$ for every dominant weight λ. So Lemma 2.1(3) implies that $\mathfrak{g}$ is $\mathbb{R}$-universal. □

References

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[2] Prasad G., Rapinchuk A., On the existence of isotropic forms of semi-simple algebraic groups over number fields with prescribed local behavior, *Adv. Math.* **207** (2006), 646–660.

[3] Tits J., Représentations linéaires irréductibles d'un groupe réductif sur un corps quelconque, *J. Reine Angew. Math.* **247** (1971), 196–220.